F78AA1 Actuarial and Financial Mathematics A

Feedback on December 2008 Examination

The following comments cover most of the features that emerged consistently in the December 2008 examination. The examination paper is available on Vision.

1. Part (a) was generally either done well, or no serious attempt was made. Most students gained some marks in part (b), and many full marks. The item done least well was (iii), often because the correct formula $(1-v^n)/d^{(4)}$ was used and then $d^{(4)}$ was calculated incorrectly. It is probably more straightforward just to change the time unit.

2. Part (a) was generally done well; nearly all students used a correct methodology and errors, if any, were in the precise timings of the cashflows. A surprising number of students found the accumulation in part (b) by direct calculation from the cashflows, instead of applying the appropriate accumulation factor to the result from part (a) — this represents wasted time.

3. Part (a) was done well by nearly everyone, very occasionally an arithmetic error crept in. Part (b) was also done well. Most students formulated the problem correctly, although many stated that:

$$\frac{\text{capital outstanding}}{a_{\overline{n}|}^{0.007}} = \text{current level repayment}$$

   when an inequality ($\leq$) is appropriate (because n is an integral number of months). However this did not prevent most from going on to find the correct extension of the term.

4. The majority of students set up a correct expression for the bond price, although some then made arithmetic errors in evaluating it. Otherwise, the most common errors were: failing to *check* or *verify* that CGT was payable; or using a redemption amount of 100 instead of 98.

5. Part (a) was very well done, and part (b) was nearly as good. The great majority of students found the IRR (yield) correctly, although one common error was in the linear interpolation, obtaining correct evaluations of $f(0.19)$ and $f(0.20)$ but then finding an interpolated value close to 20% when it should have been very nearly 19%. Part (c) was very badly done with few correct answers. The key point is to find the time when the load is repaid, at which point the interest rate switches from 6% on borrowings to 4% on savings. Most students tried to write down a present value or accumulation for the whole five year period, which cannot be done.

6. Parts (a) and (b) were generally well done by those attempting the question. The number of non-attempts may be explained by this topic being the last to be covered in the lectures, but *something* will always have to be the last topic covered in the lectures, and will certainly be examinable. Part (c) was undoubtedly an intricate calculation, and it was good to see a number of correct answers; however many students did not attempt it, or got a bit lost in setting up the equation of value. Part (d) was answered quite well, even by many who did not finish part (c).

7. This whole question was done well by most who attempted it. Quite a number of students tackled (a) by trial-and-error and linear interpolation, which is not wrong, but since the equation of value is quadratic, direct solution is much quicker.